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$$\int \frac{2x(x+5)}{x^4-1} dx = \int \frac{2x^2+10x}{(x-1)(x+1)(x^2+1)} dx = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

$$A = \left. \frac{2x^2+10x}{(x+1)(x^2+1)} \right|_{x=1} = 3$$

$$B = \left. \frac{2x^2+10x}{(x-1)(x^2+1)} \right|_{x=-1} = 2$$

$$Cx+D \Rightarrow \frac{2i^2+10i}{(i-1)(i+1)} = 1-5i \Rightarrow C = -5 \text{ en } D = 1$$

$$\Rightarrow 3 \int \frac{1}{x-1} dx + 2 \int \frac{1}{x+1} dx + \int \frac{-5x+1}{x^2+1} dx$$

$$= 3 \ln|x-1| + 2 \ln|x+1| - 5 \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx$$

$$= 3 \ln|x-1| + 2 \ln|x+1| - \frac{5}{2} \ln|x^2+1| + \text{Bgtan}(x) + C$$

References